

STARE: Spend To Acquire Reliable Evidence for Risk-Controlled Target Selection and Navigation on Edge UAVs

Anonymous submission

Abstract

Edge UAVs tasked with instructed target selection commit to a commanded target (e.g., “tree/house/car”) from a monocular camera under tight compute, time, and motion budgets, where a premature wrong commitment can forfeit the mission. Existing risk-aware methods scale perception, withhold decisions, or defer to a supervisor; none spends the agent’s own motion, which genuinely reshapes the observation distribution, under a guarantee on commit risk. We propose STARE (Spend To Acquire Reliable Evidence), an agent that treats reliability as a resource acquired by spending: a learned reliability probe and a wealth accumulator form an evidence state, a Lagrangian-PPO policy decides whether further evidence is worth its cost under a budget, and commit thresholds calibrated by Learn-then-Test on end-to-end rollouts give finite-sample, distribution-free control of the error-commit risk. Across six aerial datasets, four frozen depth backbones, and three target categories, STARE raises end-to-end success by up to 0.52 absolute over look-once commitment while keeping every reported selective-error estimate below its calibrated target; the certificate can be recalibrated on target-domain episodes, and the same agent on an edge multirotor commits correctly on real outdoor flights while abstaining where evidence stays thin.

Introduction

Instructed target selection and navigation gives a UAV *fly to the $\langle class \rangle$ at commanded coordinate w* from an upstream mission and asks it to pick the correct instance among visually similar candidates with only a monocular camera and a low-power edge processor. A natural zero-shot pipeline pairs open-vocabulary detection (Liu et al. 2024; Cheng et al. 2024) with foundation monocular depth (Yang et al. 2024; Bochkovskiy et al. 2025; Hu et al. 2024), supporting inspection, search and rescue, and aerial visual-language navigation (Liu et al. 2023; Lee et al. 2024; Chen et al. 2026; Zhang et al. 2026b). Once several plausible candidates are detected, the bottleneck moves from perception to commitment: every inference costs compute, every motion spends mission time, and a wrong commitment may exhaust the budget needed to recover.

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Existing risk-aware methods differ in what they spend when uncertainty is high. Larger depth models improve average accuracy (Yang et al. 2024; Bochkovskiy et al. 2025; Hu et al. 2024; Kong et al. 2023) but preserve the commitment errors of aerial domain shift at higher cost (Florea and Nedevschi 2025; Si et al. 2026). Conformal tools bound prediction risk (Angelopoulos and Bates 2023; Angelopoulos et al. 2025; Bates et al. 2021; Hultberg, Zachariah, and Ribeiro 2026) yet answer high uncertainty by withholding the decision (Wang et al. 2026) or deferring to a supervisor (Ren et al. 2023; Liang, Zhang, and Fisac 2024; Lekeufack et al. 2024; Gonzales et al. 2025); in flight, terminal abstention forfeits the mission and no supervisor is available (Lindemann et al. 2023). Active perception spends motion or borrowed viewpoints to gather evidence (Jin et al. 2023; Placed et al. 2023; Podolinsky et al. 2026; Fan et al. 2024) but treats motion as unbudgeted exploration without distribution-free control of the final commitment (Parr, Pezzulo, and Friston 2022). No existing agent spends its own motion, which genuinely reshapes the observation distribution, under an end-to-end guarantee on commit risk (Dao et al. 2026; Ye et al. 2026).

We propose STARE (Spend To Acquire Reliable Evidence), which treats each commitment as a hypothesis to be verified: below the calibrated commit threshold the agent keeps spending—observations, perception refreshes, motion toward informative viewpoints—until its closed-loop decision rule meets the risk target.

Our contributions are threefold:

- **Verify by spending.** The *Cascaded Evidential Introspection Module* (CEI) formulates edge-UAV target commitment as risk-controlled spend-to-verify: a reliability probe cascaded with an evidence accumulator turns uncertain perception into commit-or-defer decisions and abstention into a signal to seek evidence.
- **Spend by learning.** The *Budget-Constrained Evidence Allocation Module* (BEA) trains a PPO stopping policy that decides whether further evidence is worth its cost under a budget constraint.
- **Learn with guarantees.** The *Distribution-Free Risk Certification Module* (DRC) establishes finite-sample, distribution-free control of the error-commit risk via Learn-then-Test calibration of the dual-channel commit gate, with group-conditional extensions and cross-domain

STARE INFERENCE LOOP · VERIFY BY SPENDING

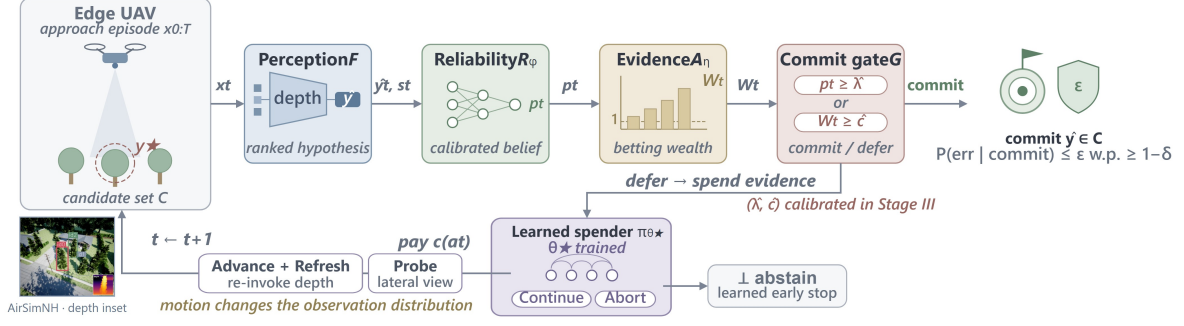


Figure 1: The STARE inference loop. **CEI** cascades $\mathcal{F} \rightarrow \mathcal{R}_\phi \rightarrow \mathcal{A}_\eta$, lifting frame x_t into calibrated belief p_t and wealth W_t ; **DRC** authorizes **COMMIT** only when $p_t \geq \hat{\lambda}$ or $W_t \geq \hat{c}$ (guarantee of Eq. (12)); on **DEFER**, **BEA** decides via π_{θ^*} whether further evidence is worth its cost, paying $c(a_t)$ to re-enter the loop.

recalibration.

Related Work

Risk-Controlled Decision-Making for Embodied Agents

Conformal prediction and its risk-control generalizations (Angelopoulos and Bates 2023; Bates et al. 2021; Angelopoulos et al. 2025) provide finite-sample, distribution-free guarantees, extended to anytime-valid, non-monotone risks (Hultberg, Zachariah, and Ribeiro 2026) and group-conditional coverage (Ding et al. 2023). In robotics they underpin abstain-then-ask deferral (Ren et al. 2023; Liang, Zhang, and Fisac 2024) and safe planning under prediction error (Lindemann et al. 2023; Lekeufack et al. 2024; Gonzales et al. 2025); risk control has reached decision policies (Joshi et al. 2026) and agentic pipelines (Opoku and Banahene 2026). Prior methods calibrate a fixed predictor and treat abstention as terminal; STARE certifies the closed-loop behavior itself, whose commit events exist only through the learned spender’s trajectory and are invisible to per-prediction calibration, and low evidence triggers further acquisition rather than surrender.

Test-Time Resource Allocation for Reliable Perception and Action

Scaling test-time compute improves reasoning reliability in language models (Snell et al. 2024; Guo et al. 2025; Shao et al. 2024); embodied instantiations allocate it in planners (Dao et al. 2026), control (Chun et al. 2026), and manipulation (Ye et al. 2026). Motion is likewise spendable: active perception and next-best-view planning have long held this view (Placed et al. 2023; Jin et al. 2023), sequential testing gives it an optimal-stopping form (Parr, Pezzulo, and Friston 2022), and recent variants borrow a teammate’s viewpoint under per-prediction conformal guarantees (Podolinsky et al. 2026) or select views by evidential uncertainty (Fan et al.

2024), leaving the final decision without distribution-free control. Agentic UAV systems interleave perception, reasoning, and movement under tight onboard resources (Zhang et al. 2026b,a; Chen et al. 2026), with constrained reinforcement learning (Schulman et al. 2017; Gu et al. 2024) as the optimization machinery. STARE treats time, computation, and motion as one budgeted evidence currency and couples stopping with a distribution-free guarantee on commit risk.

Method

STARE turns uncertain monocular perception into risk-controlled commitment by composing three modules, CEI, BEA, and DRC, into the closed loop of Fig. 1. Given observation stream $x_{0:T}$, waypoint w , and candidate set $\mathcal{C} = \{o_1, \dots, o_K\}$, the agent outputs a committed target or a deliberate abstention,

$$\hat{o} = \left(\underbrace{\mathcal{G}_{\hat{\lambda}, \hat{c}}}_{\text{commit gate}} \circ \underbrace{\pi_{\theta^*}}_{\text{learned spender}} \circ \underbrace{\mathcal{A}_\eta \otimes \mathcal{R}_\phi}_{\text{evidence state}} \circ \underbrace{\mathcal{F}}_{\text{perception}} \right) (x_{0:T}) \quad (1)$$

where $\hat{o} \in \mathcal{C} \cup \{\perp\}$ with $\perp$ abstention: $\mathcal{F}$ lifts each frame into a ranked hypothesis, $\mathcal{R}_\phi$ and $\mathcal{A}_\eta$ form the evidence state, π_{θ^*} decides whether further evidence is worth its cost, and $\mathcal{G}_{\hat{\lambda}, \hat{c}}$, the only module authorized to commit, is calibrated so the *end-to-end* risk of (1) is controlled.

Problem Formulation

At each step $t \in \{0, \dots, T-1\}$ the agent selects $a_t \in \{\text{COMMIT}\} \cup \mathcal{A}_{\text{acq}}$ with $\mathcal{A}_{\text{acq}} = \{\text{ADVANCE+REFRESH}, \text{PROBE}, \text{ABORT}\}$: **ADVANCE+REFRESH** continues the approach and re-invokes depth from a new viewpoint, **PROBE** buys an auxiliary lateral view, **ABORT** abstains early. Each acquisition carries a measured cost $c(a_t) = c_{\text{cmp}} + c_{\text{time}}(a_t) + c_{\text{mot}}(a_t)$, accumulated as $C_{\text{ep}} = \sum_t c(a_t)$.

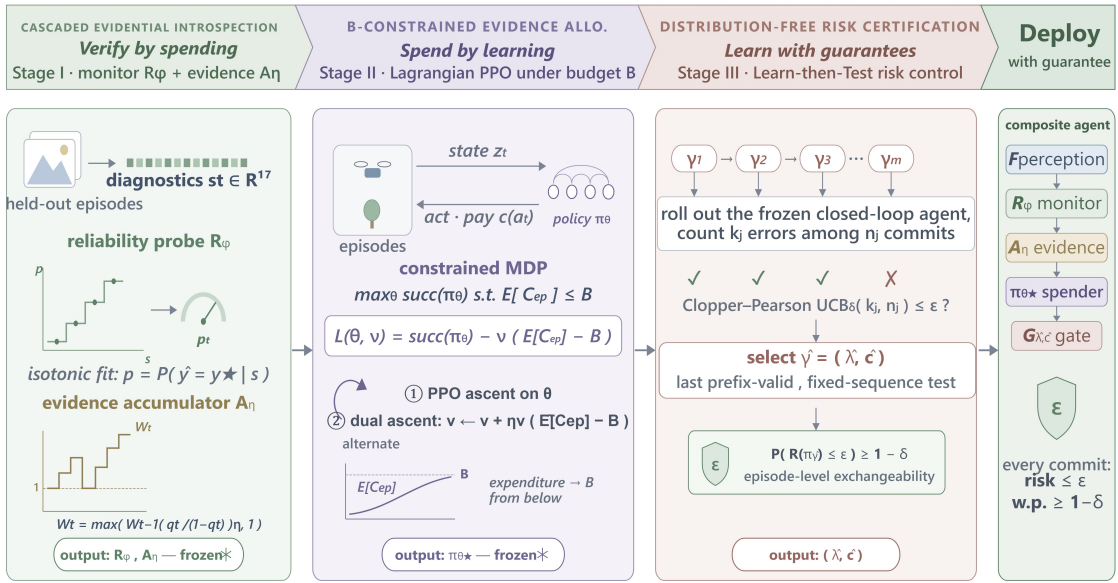


Figure 2: STARE training pipeline; module boxes match Fig. 1. **Stage I** fits the reliability probe $\mathcal{R}_\phi$ and sets the wealth accumulator $\mathcal{A}_\eta$; **Stage II** trains the spender π_{θ^*} by Lagrangian PPO on Eq. (9); **Stage III** calibrates $(\hat{\lambda}, \hat{c})$ by Learn-then-Test on end-to-end rollouts, yielding $\mathbb{P}(R(\pi_{\hat{\gamma}}) \leq \varepsilon) \geq 1 - \delta$.

With commit the event that (1) returns $\hat{o} \neq \perp$, err the event $\hat{o} \neq y^*$, and $\tilde{y}$ the effective output ($\hat{o}$ on commit; the final top-1 hypothesis $\hat{y}_T$ on abstention, so declining to act is never rewarded), we define the error-commit risk, coverage, and success rate

$$\begin{aligned} R(\pi) &= \mathbb{P}(\text{err} \mid \text{commit}), \\ \text{cov}(\pi) &= \mathbb{P}(\text{commit}), \\ \text{succ}(\pi) &= \mathbb{P}(\tilde{y} = y^*), \end{aligned} \quad (2)$$

and pose target commitment as the constrained program

$$\begin{aligned} \max_{\pi} \text{cov}(\pi) \quad \text{s.t.} \quad & \mathbb{P}(R(\pi) \leq \varepsilon) \geq 1 - \delta, \\ & \mathbb{E}[C_{\text{ep}}] \leq B, \end{aligned} \quad (3)$$

so abstention is the *interior* of the feasible region, escaped by spending: STARE buys back the largest coverage the budget affords (training pipeline in Fig. 2).

Cascaded Evidential Introspection Module: Verify by Spending

CEI is the inner cascade $(\mathcal{A}_\eta \otimes \mathcal{R}_\phi) \circ \mathcal{F}$ of (1); one pass is one decision step.

(a) Hypothesis Lift $\mathcal{F}$: pixels $\rightarrow$ ranked hypothesis. The perception operator maps frame x_t to the top-ranked candidate and deployment-observable diagnostics,

$$\mathcal{F}: x_t \mapsto (\hat{y}_t, s_t), \quad \hat{y}_t = \arg \min_{k \in [K]} \rho_t(o_k; w), \quad (4)$$

where ρ_t scores each candidate by combining bearing error with a depth-based distance residual (metric backbones rank by $\|\hat{p}_{t,k} - w\|$; relative-depth backbones break along-ray ambiguity with depth order), and $s_t \in \mathbb{R}^{16}$ collects the top-two association margin, waypoint pixel distance, cross-frame association stability, closing-rate agreement, and detection geometry.

(b) Reliability Probe $\mathcal{R}_\phi$: hypothesis $\rightarrow$ calibrated belief. Raw depth confidence is miscalibrated under aerial shift, so the diagnostics s_t are mapped to a calibrated posterior

$$\mathcal{R}_\phi: s_t \mapsto p_t = \mathbb{P}(\hat{y}_t = y^* \mid s_t) \in [0, 1], \quad (5)$$

trained on held-out episodes with isotonic calibration.

(c) Evidence Accumulator $\mathcal{A}_\eta$: per-frame belief $\rightarrow$ accumulated evidence. A single frame may be unconvincing while a streak of concordant frames is decisive, so p_t is folded into a betting-inspired wealth,

$$\begin{aligned} \mathcal{A}_\eta: (p_t, W_{t-1}) &\mapsto W_t = \max\left(W_{t-1} \left(\frac{q_t}{1-q_t}\right)^\eta, 1\right), \\ q_t &= \text{clip}(p_t; \rho, 1-\rho), \quad W_0 = 1, \end{aligned} \quad (6)$$

where each frame multiplies the wealth by a tempered likelihood ratio, $\eta < 1$ discounts temporal correlation, and the unit floor lets only uninterrupted runs of positive evidence clear the bar. W_t is purely an evidence-shaping statistic; the certificate never rests on its form, since DRC calibrates the deployed agent end-to-end.

(d) Commit Gate $\mathcal{G}_{\lambda,c}$: accumulated evidence $\rightarrow$ commit-or-defer. The signals (p_t, W_t) enter a dual-channel gate,

$$\mathcal{G}_{\lambda,c}: (p_t, W_t) \mapsto g_t = \begin{cases} \text{COMMIT} & \text{if } p_t \geq \lambda \text{ or } W_t \geq c, \\ \text{DEFER} & \text{otherwise,} \end{cases} \quad (7)$$

whose thresholds (λ, c) are calibrated below; since $c = \infty$ recovers the single-channel gate, the design only adds coverage. On DEFER the agent spends and re-enters (a) at $t+1$ with a sharper observation distribution.

Budget-Constrained Evidence Allocation Module: Spend by Learning

The gate decides when the agent *may* commit; BEA decides, on defer, whether spending is *worth it* and which acquisition to buy, since budget spent on unrecoverable episodes is taken from recoverable ones. Spending is a constrained MDP over the evidence state $z_t = (p_t, W_t, \text{margin}_t, B - \sum_{s \leq t} c(a_s), \dots)$; the spender, a light temporal-attention policy, reads the trailing H -step trace and sees the shape of the evidence trajectory rather than one compressed step,

$$\pi_\theta : z_{t-H+1:t} \mapsto a_t \in \mathcal{A}_{\text{acq}} = \{\text{ADVANCE+REFRESH, PROBE, ABORT}\}, \quad (8)$$

where PROBE is offered only on splits with a lateral branch and commit authority stays exclusively with the gate, so learning reshapes *where* budget is spent without touching *what* risk is tolerated. The spender solves

$$\theta^* = \arg \max_{\theta} \text{succ}(\pi_\theta) \quad \text{s.t.} \quad \mathbb{E}_{\pi_\theta}[C_{\text{ep}}] \leq B, \quad (9)$$

optimized as the Lagrangian saddle point

$$\theta^* = \arg \max_{\theta} \min_{\nu \geq 0} \mathcal{L}(\theta, \nu) = \text{succ}(\pi_\theta) - \nu(\mathbb{E}_{\pi_\theta}[C_{\text{ep}}] - B), \quad (10)$$

alternating PPO ascent on θ with dual ascent $\nu \leftarrow \max(0, \nu + \eta_\nu(\mathbb{E}[C_{\text{ep}}] - B))$.

Distribution-Free Risk Certification Module: Learn with Guarantees

DRC calibrates the commit gate $(\hat{\lambda}, \hat{c})$ by Learn-then-Test on end-to-end rollouts: standard conformal pipelines calibrate a *prediction*, STARE a *behavior*. Commit events arise from a nonlinear interaction of perception, gating, and policy, so the risk in (3) is non-monotone in (λ, c) and we treat $\gamma = (\lambda, c)$ as policy hyperparameters. Over a grid $\gamma_1 \succ \dots \succ \gamma_m$ from most to least conservative and n calibration episodes, we roll out the full closed-loop agent

$$\pi_{\gamma_j} = \mathcal{G}_{\gamma_j} \circ \pi_{\theta^*} \circ (\mathcal{A}_\eta \otimes \mathcal{R}_\phi) \circ \mathcal{F}, \quad (11)$$

count k_j errors among n_j commits, and select

$$\hat{\gamma} = \gamma_j, \quad \hat{j} = \max \left\{ j : \overline{\text{UCB}}_\delta(k_j, n_j) \leq \varepsilon \quad \forall i \leq j \right\}, \quad (12)$$

where $\overline{\text{UCB}}_\delta$ is the Clopper–Pearson upper bound and the prefix structure implements fixed-sequence testing, valid for non-monotone risks without multiplicity correction.

Risk control for the closed-loop agent. For calibration and deployment episodes drawn from the same domain, (12) satisfies $\mathbb{P}(R(\pi_{\hat{\gamma}}) \leq \varepsilon) \geq 1 - \delta$ over the draw of the calibration set. The guarantee holds for any policy class, including the learned π_{θ^*} , because the tested quantity is the deployed agent’s end-to-end commit-event risk rather than any internal score.

Randomness is quantified at the episode level, so intra-episode dependence is absorbed into the rollout. Mondrian-style per-group calibration upgrades the marginal guarantee to group-conditional risk control, so hard strata commit less, and when the domain moves, re-running (12) on target-domain episodes can restore the certificate. Algorithm 1 assembles the full cascade (1).

Algorithm 1: STARE: Spend To Acquire Reliable Evidence

Input: observation stream $x_{0:T}$, commanded waypoint w , candidate set $\mathcal{C}$

Parameter: $\hat{\gamma} = (\hat{\lambda}, \hat{c})$ calibrated via (12) (level δ , target ε); spender π_{θ^*} trained via (10) (budget B); probe $\mathcal{R}_\phi$

Output: committed target $\hat{o} \in \mathcal{C}$ or abstention $\perp$

```

1:  $W \leftarrow 1$ 
2: for  $t = 0, 1, \dots, T - 1$  do
3:    $(\hat{y}_t, s_t) \leftarrow \mathcal{F}(x_t)$ ;  $p_t \leftarrow \mathcal{R}_\phi(s_t)$ ;  $W \leftarrow \mathcal{A}_\eta(p_t, W)$ 
    $\triangleright$  Eqs. (4)–(6)
4:   if  $p_t \geq \hat{\lambda}$  or  $W \geq \hat{c}$  then
5:     return  $\hat{y}_t$   $\triangleright$  gate (7); risk  $\leq \varepsilon$  w.p.  $\geq 1 - \delta$ 
6:   else if  $\pi_{\theta^*}(z_{t-H+1:t}) = \text{ABORT}$  then
7:     return  $\perp$   $\triangleright$  learned early stop (8)
8:   else
9:     execute  $a_t = \pi_{\theta^*}(z_{t-H+1:t}) \in \{\text{ADVANCE+REFRESH, PROBE}\}$ ; pay  $c(a_t)$ 
10:  end if
11: end for
12: return  $\perp$ 

```

Experiment

Task Design

Every dataset instantiates one instruction template, *fly to the $\langle \text{class} \rangle$ at commanded coordinate w* , with w a noisy waypoint from an upstream planner. Given an approach episode with stream $x_{0:T}$, waypoint w , and candidate set $\mathcal{C}$ of detected same-class instances tracked across frames, the agent must commit to the instance whose true world position is closest to w ,

$$y^* = \arg \min_{k \in [K]} \|P_{\text{world}}(o_k) - w\|, \quad (13)$$

or abstain. y^* is fixed for the entire episode and does not drift as the UAV moves. The agent observes the monocular stream, w , and onboard pose, never the candidates’ true 3D locations; a depth backbone ranks hypotheses via Eq. (4). The base rows of Table 1 show how hard this ranking is from single views: even the larger backbones resolve only 44%–60% of the tree episodes.

Datasets

We evaluate on six datasets spanning three candidate categories (trees, houses, vehicles), all with metric depth ground truth: **MidAir** (Fonder and Van Droogenbroeck 2019), synthetic low-altitude drone flights with *tree* candidates (1,130 episodes); **TartanAir** (Wang et al. 2020), photorealistic simulated forests with *tree* candidates (209 episodes); and **Air-Sim** (Shah et al. 2017), with *tree* (159), *house* (86), and *graph* (322) splits from *AirSimNH* plus a *car* split from *CityEnviron* (74).

Data Processing

All episodes share one schema: a multi-frame approach *trunk* with per-frame RGB, metric depth ground truth, tracked instance boxes, camera pose, waypoint w , and label y^* . MidAir

	tree												house			car		
	MidAir			TartanAir			AirSim-tree			AirSim-graph			AirSim-house			AirSim-car		
Backbone	Succ.↑	Cov.↑	Risk↓	Succ.↑	Cov.↑	Risk↓	Succ.↑	Cov.↑	Risk↓	Succ.↑	Cov.↑	Risk↓	Succ.↑	Cov.↑	Risk↓	Succ.↑	Cov.↑	Risk↓
BASE (<i>look once and commit; never abstains</i>)																		
DAv2-S	.583	—	.417	.422	—	.578	.455	—	.545	.587	—	.413	.850	—	.150	.947	—	.053
DAv2-L	.586	—	.414	.444	—	.556	.500	—	.500	.611	—	.389	.859	—	.141	.947	—	.053
Depth-Pro	.589	—	.411	.437	—	.563	.545	—	.455	.619	—	.381	.854	—	.146	.947	—	.053
Metric3D	.589	—	.411	.444	—	.556	.508	—	.492	.603	—	.397	.884	—	.116	.930	—	.070
SEQUENTIAL BASE (<i>DAv2-S; EMA / SPRT without budget, spender, or LTT</i>)																		
EMA+ <i>p</i> -thr	.802	76.2	.176	.531	34.4	.273	.795	76.9	.267	.815	81.5	.227	.857	100	.143	.842	100	.158
SPRT/patience	.797	91.9	.177	.531	50.0	.188	.641	97.4	.368	.753	98.8	.250	.810	100	.190	.789	100	.211
STARE (Ours)																		
DAv2-S	.848	58.5	.078	.685	45.0	.082	.972	52.0	.078	.958	72.0	.048	.982	58.0	.088	.968	68.0	.022
DAv2-L	.852	56.0	.072	.710	48.0	.095	.980	55.0	.055	.945	76.0	.062	.972	55.0	.105	.970	70.0	.028
Depth-Pro	.855	60.0	.085	.675	50.0	.110	.960	48.0	.065	.930	73.0	.088	.975	60.0	.095	.985	69.0	.020
Metric3D	.862	57.5	.088	.690	52.0	.090	.948	46.0	.072	.952	75.0	.042	.975	62.0	.080	.985	65.0	.024

Table 1: Look-once base, sequential baselines, and STARE (Ours) across six datasets ($\delta=0.1$; Ours Succ. is 5-seed mean, std ≤ 0.055 across cells). Sequential rows use DAV2-S; **EMA+*p*-thr** thresholds an EMA of p_t and **SPRT/patience** is a sequential probability ratio / fixed-patience stop, neither with wealth, a learned spender, or LTT. Ours runs at the per-dataset operating ε fixed on calibration data. **Succ.**: end-to-end success ($\uparrow$), abstention charged as final top-1; **Cov.**: commit rate in % ($\uparrow$; the base never abstains, shown “—”); **Risk**: test-set selective-error estimate among commits ($\downarrow$); the certificate controls deployed risk at ε with confidence $1-\delta$ over the calibration draw. Best success per dataset in **bold**.

and TartanAir trajectories are cut into approach episodes by anchoring an open-vocabulary detection (Cheng et al. 2024; Liu et al. 2024) on the near frame and tracking the K instances backward by IoU; the waypoint task is overlaid by back-projecting instance centers with pose and metric depth, sampling a target, and setting w to its position plus GPS-like noise, with frozen depth reused across ablations. AirSim episodes are flown *de novo* with instances assigned by simulator segmentation; the graph split adds a lateral PROBE branch, the only split where probing is a first-class action.

Each dataset is partitioned by *group* (site, trajectory, or scene) into disjoint train/calibration/test pools over five seeds, with nominal ratios 50/25/25 (TartanAir 50/35/15); keeping whole groups intact yields the actual test counts below. The probe, spender, and thresholds never see the scenes they are tested on—exactly the condition the certificate is issued for—and the group-conditional extension of DRC covers residual heterogeneity. All reported metrics are test-only ($n=216/45/44/84/23/19$ for MidAir/TartanAir/AirSim-tree/graph/house/car).

Evaluation Metrics

All methods are scored at the episode level (Eq. (2)). **End-to-end success** ($\uparrow$) counts episodes whose committed target equals y^* , with abstention charged as the agent’s final top-1 hypothesis (Anderson et al. 2018), so Succ. is comparable across coverages and no method buys success by declining to act. **Guaranteed coverage** $\text{cov}@_\varepsilon$ ($\uparrow$) is the committed fraction (Geifman and El-Yaniv 2017), and **selective error risk** $\text{risk}@_\varepsilon$ ($\downarrow$) is the error rate among commits, controlled at ε with confidence $1-\delta$. Each method is reported at a single operating point (Ours: the LTT-certified ε).

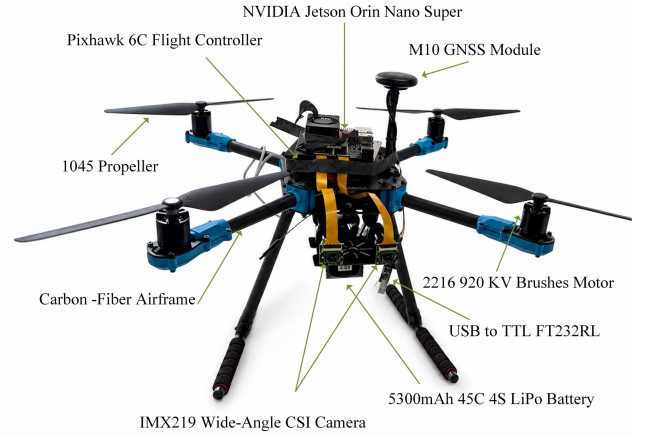


Figure 3: Edge platform: a carbon-fiber multirotor (Pixhawk 6C, M10 GNSS, IMX219 camera) with a Jetson Orin Nano Super.

Implementation

Timings are measured on an NVIDIA RTX 4080 GPU with depth cached per episode; field flights on the multirotor of Fig. 3 close Results.

CEL. The pipeline runs on four *frozen* monocular depth backbones spanning 25M–335M parameters, Depth-Anything-v2 ViT-S and ViT-L (Yang et al. 2024), Depth-Pro (Bochkovskiy et al. 2025), and Metric3D-v2 (Hu et al. 2024). $\mathcal{R}_\phi$ is a gradient-boosted tree with isotonic calibration over 16 deployment-observable sequential features, trained

	<i>tree</i>												<i>house</i>			<i>car</i>		
	MidAir			TartanAir			AirSim-tree			AirSim-graph			AirSim-house			AirSim-car		
Module	Succ.↑	Cov.↑	Risk↓	Succ.↑	Cov.↑	Risk↓	Succ.↑	Cov.↑	Risk↓	Succ.↑	Cov.↑	Risk↓	Succ.↑	Cov.↑	Risk↓	Succ.↑	Cov.↑	Risk↓
base	.583	—	.417	.422	—	.578	.455	—	.545	.587	—	.413	.850	—	.150	.947	—	.053
+ CEI	.805	42.0	.118	.640	30.0	.165	.925	35.0	.130	.920	58.0	.130	.955	48.0	.115	.955	55.0	.040
+ BEA	.838	55.0	.088	.675	42.0	.092	.960	49.0	.088	.948	68.0	.070	.972	55.0	.095	.960	64.0	.028
+ DRC	.848	58.5	.078	.685	45.0	.082	.972	52.0	.078	.958	72.0	.048	.982	58.0	.088	.968	68.0	.022

Table 2: Cumulative ablation ladder (DAv2-S backbone; $\delta=0.1$, 5-seed mean on the same test splits). Each row adds one module in order CEI $\rightarrow$ BEA $\rightarrow$ DRC. The +CEI and +BEA rungs use the same dual-channel gate with thresholds selected on the calibration pool to meet the nominal operating ε by *empirical* risk (no UCB certificate); +DRC replaces them with the LTT-calibrated pair ($\varepsilon=0.10$ on MidAir, TartanAir, AirSim-tree, AirSim-graph; 0.15 on AirSim-house, AirSim-car), isolating what finite-sample certification changes at a matched risk target.

per backbone.

BEA. π_{θ^*} is a two-block pre-LN temporal-attention encoder over the last $H=8$ evidence states (width 128, 4 heads, FFN 256; $\sim 0.3M$ parameters) with masked action logits and a value head, trained by Lagrangian PPO (Schulman et al. 2017) on the constrained MDP of Eq. (9) over the train pools (991 episodes, one spender per split, 400 PPO iterations on cached depth). Each cost $c(a_t)$ sums per-frame depth latency, inter-decision frame-wait, and probe motion; the budget is swept at quarter, half, and full spend (the main operating point).

DRC. $\hat{\gamma} = (\hat{\lambda}, \hat{c})$ is calibrated by Learn-then-Test (Angelopoulos et al. 2025) on end-to-end rollouts of the frozen composite agent, using Clopper–Pearson at $\delta = 0.1$ and a fixed-sequence prefix over the threshold grid (Eq. (12)). The operating level ε is fixed on calibration rollouts alone, before any test episode is scored, as the tightest level in $\{0.05, 0.10, 0.15, 0.20\}$ whose certified coverage exceeds 5%; this selects $\varepsilon=0.10$ on MidAir, AirSim-tree, and AirSim-graph and $\varepsilon=0.15$ on AirSim-house and AirSim-car; TartanAir selects 0.10 with DAv2-S/Metric3D and 0.15 with DAv2-L/Depth-Pro.

Results

Table 1 contrasts the look-once **base** agent with STARE; Table 2 unfolds the gain into a cumulative module ladder.

Verify by Spending

The base rows make the failure of look-once commitment concrete: selective error sits at 0.38–0.58 on every tree split, and scaling perception from the 25M-parameter DAv2-S to the 335M-parameter backbones moves base success by at most 0.09, so the bottleneck is commitment, not perception.

STARE lifts end-to-end success to 0.68–0.98 on the tree splits (+0.24–0.52 over the corresponding look-once base) and 0.97–0.99 on house and car, with every reported selective-error estimate below the corresponding operating ε . The sequential baselines stay below STARE on every dataset with selective error uncontrolled at 0.14–0.37, so temporal pooling alone replaces none of the wealth, spender, and LTT stages. With DAv2-S, guaranteed coverage ranges from

45.0% on TartanAir to 72.0% on AirSim-graph while selective error falls by roughly one-half or more relative to the sequential baselines. Since Succ. charges abstention as final top-1, these gains are not obtained merely by declining to act. Risk cells are test-set estimates at the LTT operating point; the certificate binds the deployed commit risk at confidence $1 - \delta$ over the calibration draw.

Ablation Studies

Table 2 attributes the gain to mechanism by switching on CEI, BEA, and DRC cumulatively on the shared DAv2-S backbone. CEI alone lifts success sharply over base (0.583 to 0.805 on MidAir), which is where prior selective pipelines stop, yet without a learned spender the gate leaves selective error above the operating ε on the harder tree splits (0.118–0.165). BEA buys coverage and success together at a matched empirical- ε threshold (58.0% to 68.0% coverage on AirSim-graph, 0.925 to 0.960 success on AirSim-tree) for ~ 0.3 –1.6 s of wall-clock per depth frame, within a Jetson-class budget. DRC does not renegotiate accuracy: success moves by at most +0.012 over +BEA, while the LTT certificate further reduces selective error (0.070 to 0.048 on AirSim-graph) and adds certified coverage (55.0% to 58.5% on MidAir). Sweeping ε moves coverage monotonically while success stays essentially flat: coverage dials without renegotiating accuracy.

Field Deployment

We port the complete stack on DAv2-S depth to the multirotor of Fig. 3, running onboard the Jetson Orin Nano Super with w from surveyed GNSS positions of target trees, and flew 20 field episodes over five park scenes with dense same-class candidates (Fig. 5). The agent committed in 18 episodes, all correct under post-hoc annotation, and abstained in 2 under heavy canopy motion blur, where the gate never fired. Flight logs recorded 3.4 acquisitions per commit, and in 6 of the 18 commits evidence spent along the approach flipped an early wrong top-1 association, the signature of Fig. 4.

Limitations and Future Work

Finite calibration pools can make Clopper–Pearson certification conservative, reducing coverage rather than invalidating

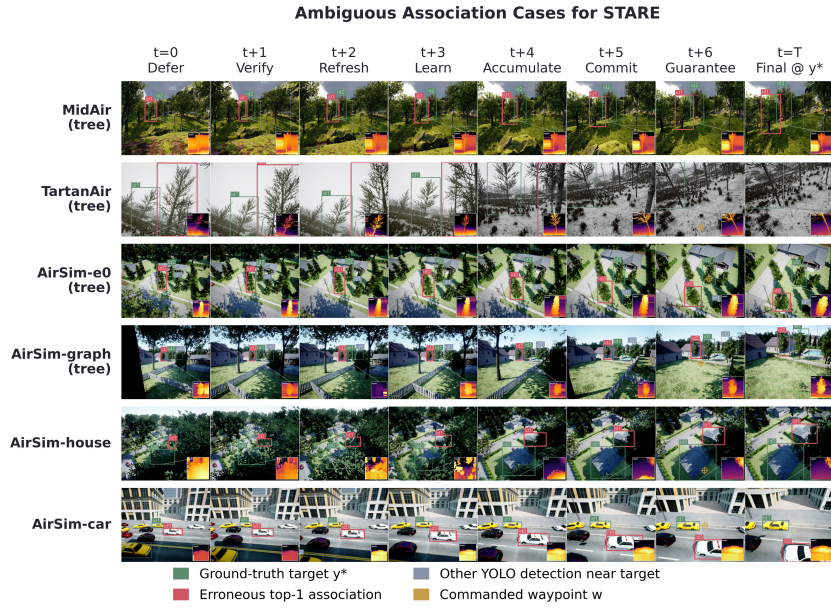


Figure 4: Ambiguous association cases across six datasets (rows) over time (columns): evidence spent along the approach flips the early wrong association, and the agent commits once the calibrated gate fires.

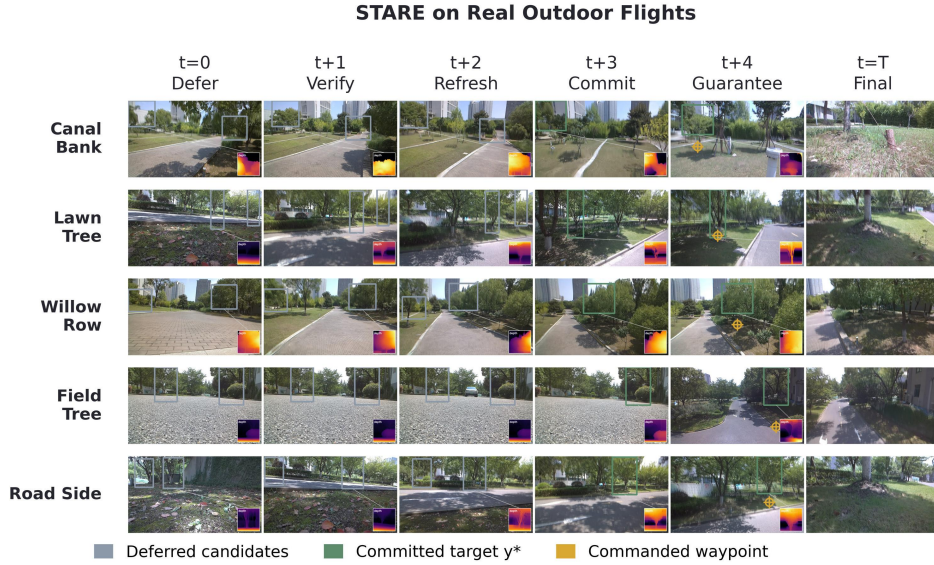


Figure 5: STARE on real outdoor flights over five park scenes (rows), with onboard depth as insets. The agent spends refreshes along the approach and commits only once the gate fires.

accepted certificates. The guarantee is distribution-specific, so substantial shifts require target-domain recalibration; this updates only commit thresholds, not the probe or spender. Future work will improve sample efficiency and extend risk control to dynamic targets and richer acquisition actions.

Conclusion

STARE treats reliability as a resource an agent buys with its own time, computation, and motion, and makes every commit carry a finite-sample, distribution-free contract. Across six

datasets, four frozen backbones, and three target categories, spending evidence raises end-to-end success while pulling selective error to the calibrated target, and the same loop on an edge multirotor commits correctly outdoors and defers when evidence never arrives. Code, episode builders, and calibration scripts will be released.

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